Chronic Pancreatitis

*By **Michael Bartel**, MD, PhD, Fairfax, VA*
*Reviewed By **Minhhuyen Nguyen**, MD, Fox Chase Cancer Center, Temple University*
Reviewed/Revised Mar 2024 | Modified Dec 2025

Chronic pancreatitis is persistent inflammation of the pancreas that results in permanent structural damage with fibrosis and ductal strictures, followed by a decline in exocrine and endocrine function (pancreatic insufficiency). Heavy alcohol consumption and smoking cigarettes are two of the major risk factors. Abdominal pain is the predominant symptom in most patients. Diagnosis is usually made by imaging studies and pancreatic function testing. Treatment mainly includes pain control and management of pancreatic insufficiency.

(See also Overview of Pancreatitis and Acute Pancreatitis.)

Fibrosis caused by inflammation and recurrent pancreatic injury is the hallmark of chronic pancreatitis but needs to be distinguished from fibrosis caused by the aging process and diabetic pancreatopathy.

Chronic pancreatitis can cause calcification of the pancreatic parenchyma, formation of intraductal stones, or both as well as pancreatic atrophy.

Pathogenesis of Chronic Pancreatitis

The pathogenesis of chronic pancreatitis is not well understood. Several mechanisms have been proposed.

The **stone and duct obstruction theory** proposes that disease is due to ductal obstruction caused by formation of protein-rich plugs as a result of protein–bicarbonate imbalance for unknown reasons. These plugs may calcify and eventually form stones within the pancreatic ducts. If obstruction is chronic, persistent inflammation leads to fibrosis, pancreatic ductal distortion, strictures, and atrophy. After several years, progressive fibrosis and atrophy lead to loss of exocrine and endocrine function.

The **necrosis–fibrosis hypothesis** posits that repeated attacks of acute pancreatitis with necrosis are key to the pathogenesis of chronic pancreatitis. Over years, the healing process replaces the necrotic tissue with fibrotic tissue, leading to the development of chronic pancreatitis. This hypothesis has evolved into the **sentinel acute pancreatitis event** (SAPE) model. The SAPE model proposes that an initial episode of acute pancreatitis leads to a hypersensitive pancreas (1). A sensitized pancreas is more vulnerable to minor stressors (alcohol, tobacco) that can cause recurrent acute pancreatitis events, whereas similar stressors would not result in acute pancreatitis before the SAPE.

In more advanced chronic pancreatitis, neuronal sheath hypertrophy and perineural inflammation occur and may contribute to chronic pain.

Pathogenesis reference

1. Whitcomb DC: Central role of the sentinel acute pancreatitis event (SAPE) model in understanding recurrent acute pancreatitis (RAP): Implications for precision medicine. *Front Pediatr* 10:941852, 2022. doi: 10.3389/fped.2022.941852

Etiology of Chronic Pancreatitis

In the United States, about 50% of cases of chronic pancreatitis result from heavy alcohol consumption, and chronic pancreatitis is more common among men than women. However, only a minority of people with sustained alcohol exposure ultimately develop chronic pancreatitis, suggesting that there are other cofactors required to trigger overt disease. Cigarette smoking is an independent, dose-dependent risk factor for developing chronic pancreatitis (1). Both heavy alcohol consumption and smoking increase risk of disease progression, and their risks are likely additive. A large proportion of cases of chronic pancreatitis are idiopathic.

Tropical pancreatitis is an idiopathic form of chronic pancreatitis that occurs in children and young adults in tropical regions such as India, Indonesia, and Nigeria. Tropical pancreatitis is characterized by an early age of onset, large ductal calculi, an accelerated course of the disease, and an increased risk of pancreatic cancer.

Less common causes of chronic pancreatitis include genetic disorders, systemic diseases, and ductal obstruction caused by stenosis, stones, or cancer (see table Causes of Chronic Pancreatitis).

TABLE

Causes of Chronic Pancreatitis

Causes	Examples
Toxins	Alcohol
Genetic	Cationic trypsinogen gene (*PRSSI*) Serine peptidase inhibitor Kazal-type 1 (*SPINK1*) Cystic fibrosis transmembrane conductance regulator gene (*CFTR*) Chymotrypsin C mutations and other

Causes	Examples
	genetic disorders
Obstructive	Pancreatic duct stricture (traumatic, iatrogenic, anastomotic, or malignant) Mass effect due to a tumor Possibly pancreas divisum (congenital anomaly causing division of the pancreatic duct) Possibly sphincter of Oddi dysfunction
Autoimmune	Type 1, related to IgG4 disease, and type 2 autoimmune pancreatitis
Idiopathic	Tropical pancreatitis
Other	Cigarette smoking

Etiology reference

1. Yadav D, Gawes RH, Brand RE, et al: Alcohol consumption, cigarette smoking, and the risk of recurrent acute and chronic pancreatitis. *Arch Intern Med* 169:1035–1045, 2009. doi: 10.1001/archinternmed.2009.125. Clarification and additional information. *Arch Intern Med* 171(7):710, 2011. doi:10.1001/archinternmed.2011.124

Complications of Chronic Pancreatitis

When lipase and protease secretions are reduced to < 10% of normal, the patient develops malabsorption characterized by steatorrhea, the passing of greasy stools, or even oil droplets that float in water and are difficult to flush. In severe cases, undernutrition, weight loss, and malabsorption of fat-soluble vitamins (A, D, E, and K) may also occur.

Glucose intolerance may appear at any time, but overt diabetes mellitus (pancreatogenic diabetes) usually occurs late in the course of chronic pancreatitis. Patients also are at risk of hypoglycemia because pancreatic alpha cells, which produce glucagon (a counter-regulatory hormone), are lost.

Other complications of chronic pancreatitis include

- Formation of pseudocysts
- Obstruction of the bile duct or duodenum
- Disruption (disconnection) of the pancreatic duct (resulting in ascites or pleural effusion)
- Thrombosis of the splenic vein (can cause gastric varices)
- Pseudoaneurysms of arteries near the pancreas or pseudocyst

Patients with chronic pancreatitis are at increased risk of pancreatic adenocarcinoma, and this risk seems to be greatest for patients with hereditary and tropical pancreatitis.

Symptoms and Signs of Chronic Pancreatitis

Abdominal pain and pancreatic insufficiency are the primary manifestations of chronic pancreatitis. Pain can occur during the early stages of chronic pancreatitis, before development of apparent structural abnormalities in the pancreas on imaging. Pain is often the dominant symptom in chronic pancreatitis and is present in most patients. Pain is usually postprandial, located in the epigastric area, and partially relieved by sitting up or leaning forward. The pain attacks are initially episodic but later tend to become continuous.

A small subset of patients have no pain and present with symptoms of malabsorption. Clinical manifestations of pancreatic insufficiency include flatulence, abdominal distention, steatorrhea, undernutrition, weight loss, and fatigue.

Diagnosis of Chronic Pancreatitis

- Imaging studies
- Pancreatic function tests

Diagnosis of chronic pancreatitis can be difficult because amylase and lipase levels are frequently normal secondary to significant loss of pancreatic function. Diagnosis relies on clinical assessment, imaging studies, and pancreatic function tests (1).

Patients with unexplained or sustained worsening of symptoms should be evaluated for pancreatic cancer, particularly if assessment reveals a pancreatic duct stricture (2). Evaluation may include brushing of the strictures for cytology and measuring serum markers (eg, CA 19-9, carcinoembryonic antigen).

Imaging studies

In a patient with a typical history of heavy alcohol consumption and recurrent episodes of acute pancreatitis, detection of pancreatic calcification on plain x-rays of the abdomen may be sufficient. However, such calcifications typically occur late in the disease and then are visible in only about 30% of patients. CT can also be used in patients with a history of heavy alcohol consumption and in whom plain x-rays are not diagnostic.

In patients without a typical history but with symptoms suggesting chronic pancreatitis, abdominal CT is typically recommended to exclude pancreatic cancer as the cause of pain. Abdominal CT can be used to detect calcifications and other pancreatic abnormalities (eg, pseudocyst or dilated ducts) but still may be normal early in the disease.

MRI coupled with magnetic resonance cholangiopancreatography (MRCP) is now frequently used for diagnosis and can show masses in the pancreas as well as provide more optimal visualization of ductal changes consistent with chronic pancreatitis. Administration of IV secretin during MRCP increases sensitivity for detecting ductal abnormalities and also allows for functional assessment in patients with chronic pancreatitis. MRI is more accurate than CT and does not expose patients to radiation.

Endoscopic retrograde cholangiopancreatography (ERCP) is invasive and rarely used for the diagnosis of chronic pancreatitis. ERCP findings could be normal in patients with early chronic pancreatitis. ERCP should be reserved for patients who may need therapeutic intervention.

Endoscopic ultrasonography is less invasive and enables detection of subtle abnormalities in the pancreatic parenchyma and in the pancreatic duct. This imaging modality has a high sensitivity but limited specificity.

Pancreatic function tests

The most common pancreatic function tests do not detect mild to moderate exocrine pancreatic insufficiency with adequate accuracy. Late in the disease, tests of pancreatic exocrine function more reliably become abnormal.

Pancreatic function tests are classified as

- Direct: Monitor the actual secretion of pancreatic exocrine products (bicarbonate and enzymes).
- Indirect: Measure the secondary effects of the lack of pancreatic enzymes (eg, fat malabsorption).

Direct pancreatic function tests are most useful in patients who have an earlier stage of chronic pancreatitis in whom imaging studies are not diagnostic. Direct tests involve intravenous infusion of the hormone cholecystokinin to measure the production of digestive enzymes or infusion of the hormone secretin to measure the production of bicarbonate. The duodenal secretions are collected using double-lumen gastroduodenal collection tubes or an endoscope. Direct tests are cumbersome, are time-consuming, and have not been well standardized. Direct pancreatic function tests have mostly been phased out of clinical practice and are done in only a few specialized centers.

Indirect pancreatic function tests are less accurate in diagnosing earlier stages of chronic pancreatitis. These tests involve blood or stool samples. The serum trypsinogen test is an inexpensive test and is available through commercial laboratories. Very low levels of serum trypsinogen (< 20 ng/mL) are highly specific for chronic pancreatitis. A 72-hour test for stool fecal fat in patients who are following a high-fat diet is diagnostic for steatorrhea. This test is fairly reliable but cannot establish the cause of malabsorption. In other tests, fecal concentration of chymotrypsin and elastase may be decreased. The indirect tests are widely available, less invasive, inexpensive, and easier to do than the direct tests.

Diagnosis references

1. Gardner TB, Adler DG, Forsmark CE, et al: ACG Clinical Guideline: Chronic Pancreatitis. *Am J Gastroenterol* 115(3):322-339, 2020. doi: 10.14309/ajg.0000000000000535

2. Greenhalf W, Lévy P, Gress T, et al: International consensus guidelines on surveillance for pancreatic cancer in chronic pancreatitis. Recommendations from the working group for the international consensus guidelines for chronic pancreatitis in collaboration with the International Association of Pancreatology, the American Pancreatic Association, the Japan Pancreas Society, and European Pancreatic Club. *Pancreatology* 20(5):910-918, 2020. doi: 10.1016/j.pan.2020.05.011

Treatment of Chronic Pancreatitis

- Pain control
- Pancreatic enzyme supplements
- Management of diabetes
- Management of other complications

The prognosis for chronic pancreatitis is variable.

Pain control

Pain control is the most challenging task in the management of patients with chronic pancreatitis (1). First, vigorous efforts and appropriate referrals to encourage smoking cessation and alcohol abstinence should be made for patients with chronic pancreatitis in an effort to slow the disease progression as early as possible. Second, treatable complications of chronic pancreatitis such as duodenal or biliary obstruction that can cause similar symptoms should be sought. Patients should consume a low-fat (< 25 g/day) diet to reduce secretion of pancreatic enzymes. Patients with chronic pancreatitis should be educated about healthy lifestyle practices, and this should be reinforced at each visit.

Pancreatic enzyme supplementation may reduce chronic pain by suppressing the release of cholecystokinin from the duodenum, thereby reducing the secretion of pancreatic enzymes. Enzyme therapy is more likely to be successful in patients with less advanced disease, in women, and in patients with idiopathic pancreatitis than in patients with alcohol-associated pancreatitis. Although enzyme therapy is often tried because of its safety and minimal adverse effects, it may not provide substantial benefit in improving pain.

Often these measures do not relieve pain, requiring increased amounts of opioids, which increases the risk of addiction. Adjunctive pain medications, such as tricyclic antidepressants, gabapentin, pregabalin, and selective serotonin reuptake inhibitors, have been used alone or combined with opioids to manage chronic pain; results are variable. Pharmacologic treatment of pain in chronic pancreatitis is often unsatisfactory.

Glucocorticoids may be used to treat autoimmune pancreatitis (2).

Other treatment modalities include endoscopic therapy, lithotripsy, celiac plexus nerve block, and surgery.

Endoscopic therapy is aimed at decompressing a pancreatic duct obstructed by stricture, stones, or both and may provide pain relief in carefully selected patients with appropriate ductal anatomy. If there is significant stricture at the papilla or distal pancreatic duct, endoscopic retrograde cholangiopancreatography (ERCP) with sphincterotomy, stent placement, or dilation may be effective (3). Pseudocysts can cause chronic pain. Some pseudocysts can be drained endoscopically (4).

Lithotripsy (extracorporeal shock wave lithotripsy or intraductal lithotripsy) is usually needed to treat large or impacted pancreatic stones.

Percutaneous or endoscopic ultrasound-guided nerve blockade of the celiac plexus with a corticosteroid and long-acting anesthetic may provide short-term pain relief in some patients with chronic pancreatitis.

Surgical treatment may be effective for pain relief. Surgical options should be reserved for patients who have stopped using alcohol and who can manage diabetes that may be intensified by pancreatic resection. A variety of surgical options involve resection and/or decompression (5). The choice of surgical procedure depends on the anatomy of the pancreatic duct, consideration of local complications, the surgical history of the patient, and local expertise. For example, if the main pancreatic duct is dilated > 5 to 8 mm, a lateral pancreaticojejunostomy (Puestow procedure) or Partington-Rochelle modification of the Puestow procedure relieves pain in about 70 to 80% of patients who are well-selected (eg, without an inflammatory mass) (6). If the pancreatic duct is not dilated, a variation of the modified Puestow procedure called a V-plasty or Hamburg procedure can be done.

Other surgical approaches include a partial resection such as a distal pancreatectomy (for extensive disease at the tail of the pancreas), a Whipple procedure (for extensive disease at the head of the pancreas), a pylorus-sparing pancreaticoduodenectomy (similar to a Whipple procedure), a duodenum-preserving pancreatic head resection (Beger procedure), or a total pancreatectomy with autotransplantation of islets. Overall, surgical drainage is more effective than endoscopic approaches in relieving obstruction and achieving pain relief (7).

Pancreatic enzyme replacement

In patients with exocrine pancreatic insufficiency, malabsorption of fat is more severe than malabsorption of proteins and carbohydrates. Fat malabsorption also results in a deficit of fat-soluble vitamins (A, D, E, and K). Pancreatic enzyme replacement therapy (replacement of deficient hormones to treat pancreatic insufficiency) is used to treat steatorrhea (8). Various preparations are available, and a dose of 75,000 to 150,000 United States Pharmacopeia units (25,000 to 50,000 international units) lipase per meal and half that amount with snacks is needed for appropriate fat absorption. The treatment should be started at these (low) doses with subsequent titration based on clinical response. The preparations should be taken with meals. An H2 blocker or proton pump inhibitor should be given to patients taking nonenteric–coated preparations to prevent acid breakdown of the enzymes.

Favorable clinical responses include weight gain, fewer bowel movements, elimination of oil droplet seepage, increases in levels of fat-soluble vitamins, and improved well-being. Clinical response can be documented by showing a decrease in stool fat after enzyme replacement therapy. If steatorrhea is particularly severe and refractory to these measures, medium-chain triglycerides can be provided as a source of fat because they are absorbed without pancreatic enzymes and other dietary fats should be reduced proportionally. Supplementation with fat-soluble vitamins A, D, and K should be given, including vitamin E, which may minimize inflammation.

Management of diabetes

The patient should be referred to an endocrine specialist for the management of diabetes. Insulin should be given cautiously because the coexisting deficiency of glucagon secretion by alpha cells can result in unopposed and prolonged hypoglycemia, which is the hallmark of pancreatogenic diabetes (type 3c diabetes). Oral hypoglycemic drugs rarely help treat diabetes caused by chronic pancreatitis.

Treatment references

1. Gardner TB, Adler DG, Forsmark CE, et al: ACG Clinical Guideline: Chronic Pancreatitis. *Am J Gastroenterol* 115(3):322-339, 2020. doi: 10.14309/ajg.0000000000000535
2. Okazaki K, Chari ST, Frulloni L, et al: International consensus for the treatment of autoimmune pancreatitis. *Pancreatology* 17(1):1-6, 2017. doi: 10.1016/j.pan.2016.12.003
3. Kitano M, Gress TM, Garg PK, et al: International consensus guidelines on interventional endoscopy in chronic pancreatitis. Recommendations from the working group for the international consensus guidelines for chronic pancreatitis in collaboration with the International Association of Pancreatology, the American Pancreatic Association, the Japan Pancreas Society, and European Pancreatic Club. *Pancreatology* 20(6):1045-1055, 2020. doi: 10.1016/j.pan.2020.05.022
4. ASGE Standards of Practice Committee; Muthusamy VR, Chandrasekhara V, et al: The role of endoscopy in the diagnosis and treatment of inflammatory pancreatic fluid collections. *Gastrointest Endosc* 83(3):481-488, 2016. doi: 10.1016/j.gie.2015.11.027
5. Kempeneers MA, Issa Y, Ali UA, et al: International consensus guidelines for surgery and the timing of intervention in chronic pancreatitis. *Pancreatology* 20(2):149-157, 2020. doi: 10.1016/j.pan.2019.12.005
6. Bouwense SAW, Kempeneers MA, van Santvoort HC, et al: Surgery in Chronic Pancreatitis: Indication, Timing and Procedures. *Visc Med* 35(2):110-118, 2019. doi: 10.1159/000499612
7. Yang CJ, Bliss LA, Freedman SD, et al: Surgery for chronic pancreatitis: The role of early surgery in pain management. *Pancreas* 44(5):819–823, 2015. doi: 10.1097/MPA.0000000000000333
8. Whitcomb DC, Buchner AM, Forsmark CE: AGA Clinical Practice Update on the Epidemiology, Evaluation, and Management of Exocrine Pancreatic Insufficiency: Expert Review. *Gastroenterology* 165(5):1292-1301, 2023. doi: 10.1053/j.gastro.2023.07.007

Key Points

- **Recurrent attacks of acute pancreatitis may cause chronic persistent inflammation, ductal damage, and ultimately fibrosis, leading to chronic pancreatitis.**
- **Patients have episodic abdominal pain, followed later in the disease by manifestations of malabsorption.**

- Diagnosis is based on clinical presentation, imaging studies, and pancreatic function tests.
- Treatment of chronic pancreatitis principally includes pain control and management of complications, including pancreatic insufficiency.

Drug Information for the Topic